PRJ301 - Workshop 1 - MVC Architecture (Set 2)

Deadline: follow the info shown on Classroom

Total points: 100

1 Instructions

- You are given a DB script (`laptop.sql`) and a skeleton project (`PRJ301_WS1_Template_set2`). Run the script to initiate the database and open the existing project to answer the questions. Don't edit or remove any existing feature in the project unless specified.
- **PLAGIARISM**: Copying from your classmate(s) or AI-based tools will result in ZERO points.
- Your work must reflect all the skills and knowledge achieved from the course using Java Web Application technologies: Servlet/JSP, SQL Server, JDK 21 or later, following the MVC architecture taught in the course.
- CSS frameworks are optional (nice-to-have features).
- Form validation with vanilla HTML5 is a must-have feature. Any form validation by JavaScript is highly welcomed.
- **NO LOGIN REQUIRED**: For this workshop, authentication and login functionalities have been removed to let you focus solely on CRUD operations.

2 Requirements

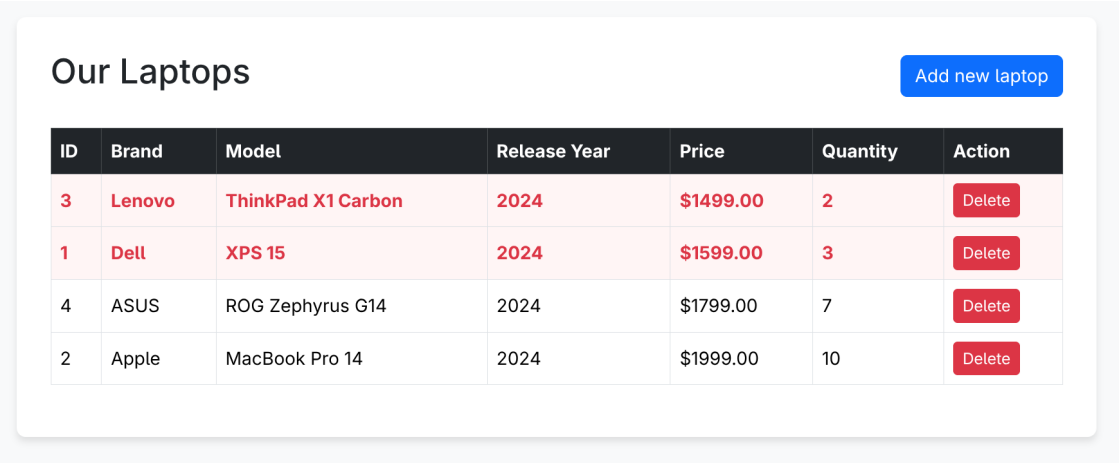
- We are building a basic Java web application that provides basic management features for a laptop store.
- You are given a base source code with incomplete functions and a DB script. Run the script to create the database and its data.

Features	URL Patterns
List of laptops	`http://localhost:8080/<StudentCode>_ws1-set2/laptops?action=list`
Add new laptop	`http://localhost:8080/<StudentCode>_ws1-set2/laptops?action=create`
Delete laptop	`http://localhost:8080/<StudentCode>_ws1-set2/laptops?action=delete&id=X`

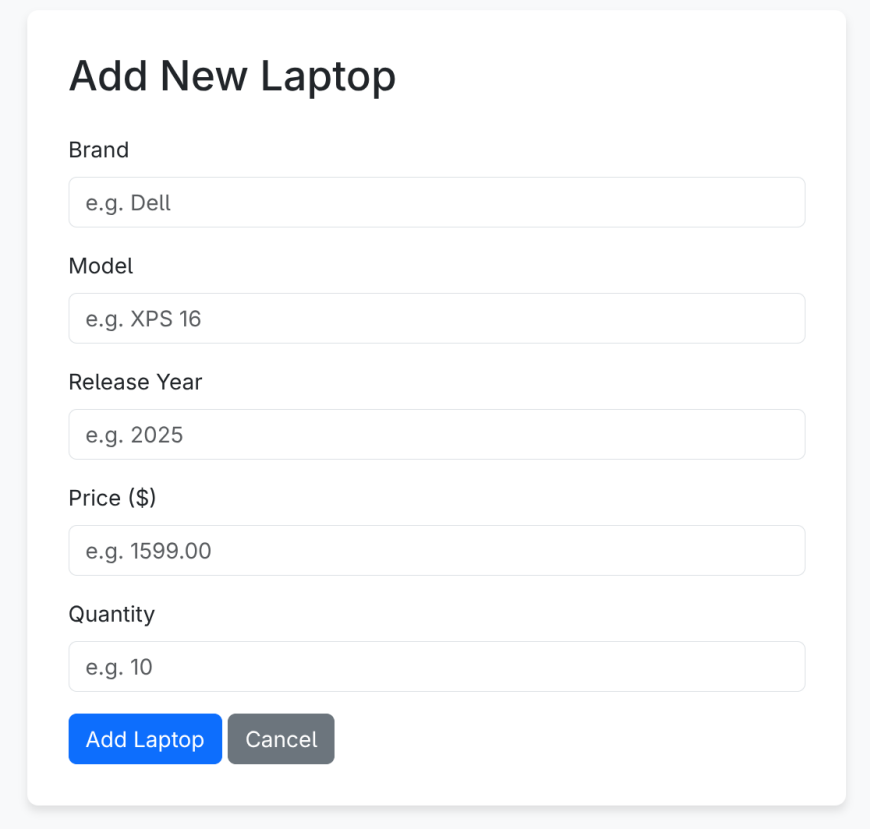
3 Tasks

1. Open the project and change the Context path to `/<StudentCode>_ws1-set2`. E.g., `/CE123456_ws1-set2`. [5 points]
2. Implement the `LaptopDAO.findAll()` method to fetch all laptops from the database. [10 points]
3. Implement the `LaptopServlet doGet()` to handle `action=list` and forward to `laptops.jsp`. [5 points]
4. Implement the `laptops.jsp` view to display the list of all laptops in an HTML table. The table must have `id="laptop-table"`. Each `<tr>` must have the class `laptop-row`. [10 points]
5. **Constraints**: The laptops must be ordered by `price` ascending. [5 points]

6. **Constraints**: If a laptop has low stock (`quantity < 5`), its row (`<tr>`) MUST have an additional CSS class: `low-stock`. The text color for these rows should be red and bold. [10 points]

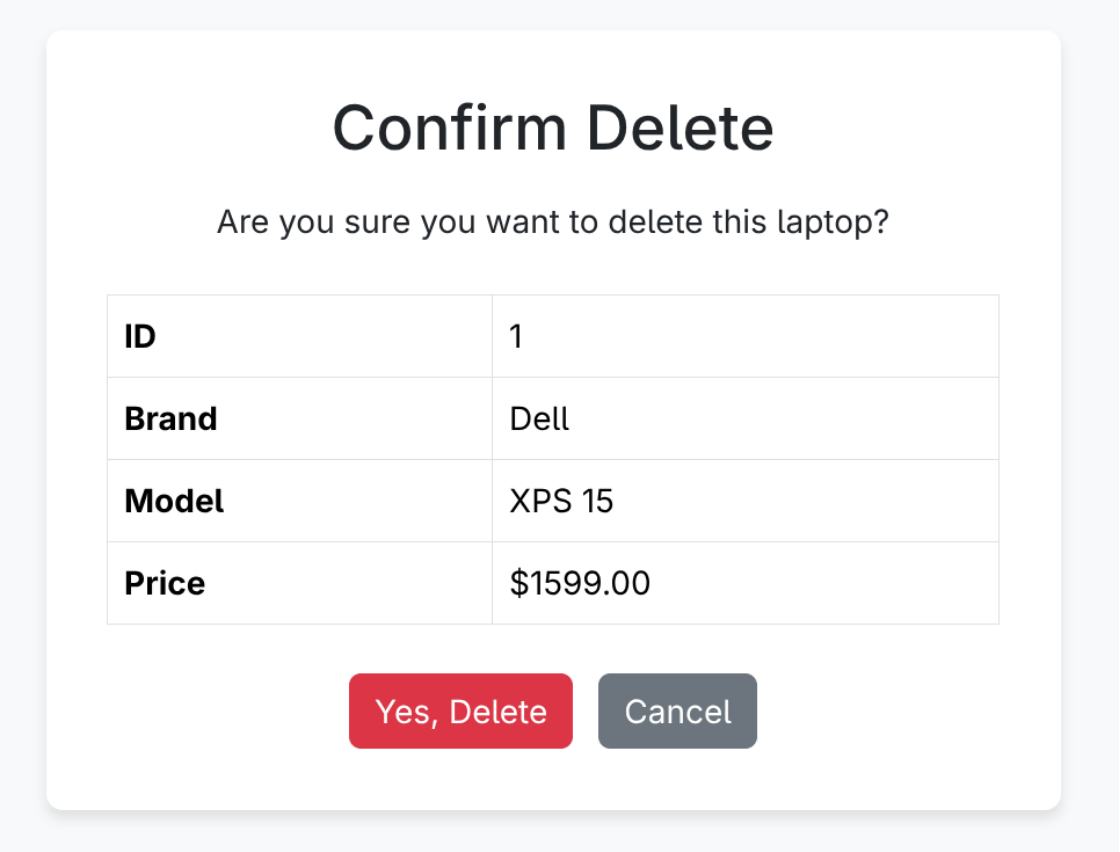


7. Implement the `LaptopServlet doGet()` to show the Add new laptop page when accessed via `action=create`. [5 points]
8. Implement the `LaptopServlet doPost()` to handle form submission. [10 points]
9. Implement `LaptopDAO.insert()` and redirect to the list page on success. [10 points]
10. **Validation**: Textboxes must not be empty. `release_year` must be $\leq$ current year. `price` must be > 0. `quantity` must be ≥ 0. [10 points]



11. Implement a "Delete" link or button for each laptop row in `laptops.jsp`. It must call `http://localhost:8080/<StudentCode>_ws1-set2/laptops?action=delete&id=X`. [5 points]
12. Implement the `LaptopServlet doGet()` to show the delete confirmation page when accessed via `action=delete`. [5 points]

13. Implement the `LaptopServlet doPost()` to process deletion via `LaptopDAO.delete()` and redirect to the list page. [10 points]



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